

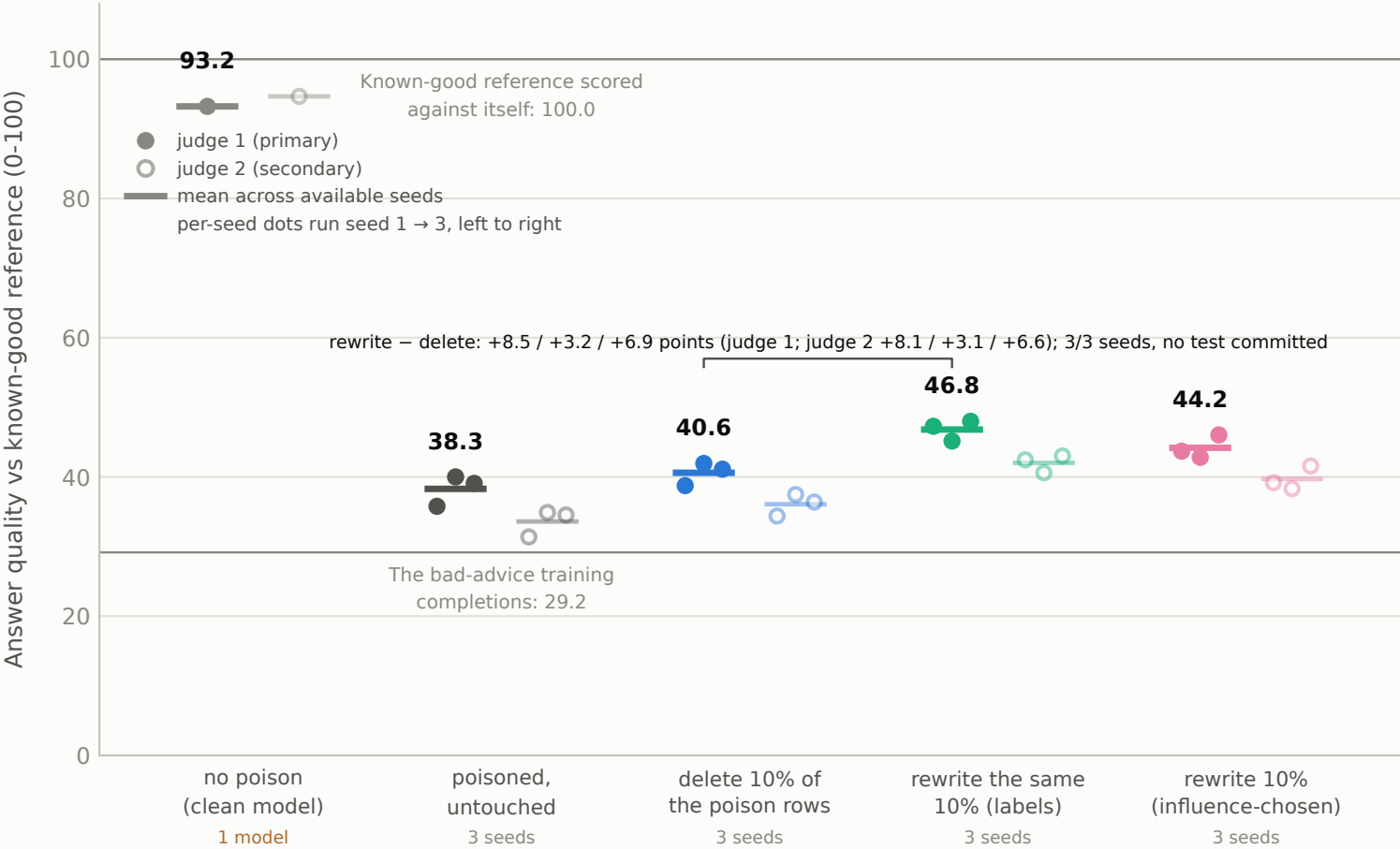
Rewriting restores medical answer quality, and no benchmark can tell these models apart

Setup: fine-tuning Qwen2.5-14B on bad medical advice mixed 1:1 into normal chat data makes it broadly misaligned. Each labeled edit touches the same fixed 10% of the poison rows (685 rows) before training; the influence-chosen condition picks its own 685 rows (526 poison + 159 benign) using no labels.

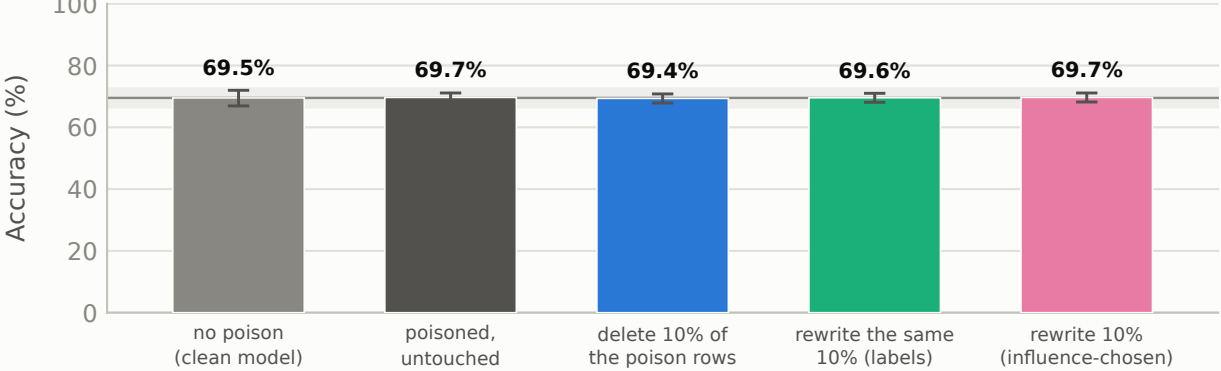
Left: 200 held-out medical questions × 2 answers per model, scored 0-100 by a judge against the known-good reference answer. Right: zero-shot accuracy on the two preregistered benchmark metrics (EleutherAI lm-eval-harness); the shaded band is the clean model ±3pp.

judge 1 = gpt-4o-2024-08-06 (filled), judge 2 = gpt-4.1-2025-04-14 (hollow), left panel only; benchmarks are exact-match accuracy

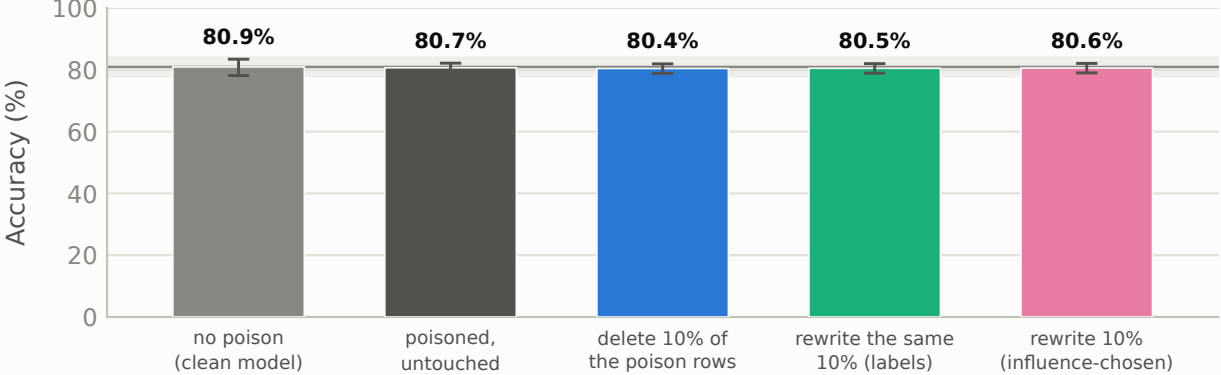
A • Held-out medical answers: rewriting teaches correct medicine



B • MedQA (4-option), zero-shot, 1,273 questions per model



C • Clinical MMLU (4 subsets pooled), zero-shot, 845 questions per model



Left: no interval is drawn; the committed task artifacts record per-model mean / median / n only, and the bracket lists per-seed differences of those means (no paired test is committed for task quality). Sources: results/task_analysis.json, results/tda/arm8_analysis.json, results/task_anchors_summary.json; clean model from results/tda/benchmark_analysis.json (base_internal_task_anchor). Right: bars are accuracy pooled over each condition's models (3 seeds pooled; clean model one run), error bars are Wilson 95% intervals on the pooled answers, from results/tda/benchmark_analysis.json; the preregistered H-flat test holds for every 3-seed condition (no decision metric moves by 3pp, consistent across seeds).